

Multiple Choice

1. **Answer:** A

Options: A) 6; B) 9; C) 2; D) 4; E) 10

Note: Correct option: A - 6

2. **Answer:** D

Options: A) 0.175 nm; B) 0.080 nm; C) 0.098 nm; D) 0.123 nm; E) 0.135 nm

Note: Correct option: D - 0.123 nm

3. **Answer:** A

Options: A) 200 Å; B) 1 Å; C) 0.1 Å; D) 50 Å; E) 500 Å

Note: Correct option: A - 200 Å

4. **Answer:** A

Options: A) CH₃COOH; B) CO₂; C) O₂; D) H₂O; E) NH₃

Note: Correct option: A - CH₃COOH

5. **Answer:** D

Options: A) Silicon; B) Boron-doped silicon; C) Pure germanium; D) Arsenic-doped silicon; E) Indium Phosphide

Note: Correct option: D - Arsenic-doped silicon

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '17 O'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '- 275 J'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '-2'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '15'.

Long Form Response

11. **Answer:** 36 mJ. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 92.22 MeV and 7.68 MeV. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.